

Review III

1. Section 8.1

- a. Sequence $\{a_n\}$
 - i. General term a_n
 - ii. Function $a_n = f(n)$
- b. Limit of Sequence
 - i. Definition
 - ii. Graphical Interpretation
 - iii. Language: Converge vs Diverge
 - iv. Application of Definition: $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$
- c. Limit Theorems
 - i. Linearity (Sum, Difference, Constant Multiple)
 - ii. Product, Quotient, Functional Composition
- d. Definition $\lim_{n \rightarrow 0} a_n = \infty$
- e. Sandwich Theorem for Sequences
- f. Monotone Sequences
 - i. Increasing
 - ii. Decreasing
- g. Bounded Sequences
 - i. Bounded Above
 - ii. Bounded Below
 - iii. Bounded
- h. Bounded Monotone Convergence Theorem
- i. Limits of 6 Commonly Occurring Sequences
- j. Representative Problems: 12, 16, 20, 23, 25, 26, 28, 29, 31, 33, 34, 36, 37, 40, 42, 44, 49, 50

2. Section 8.2

- a. Infinite Series $\sum_{n=1}^{\infty} a_n$
 - i. Terms of the Series: a_n
 - ii. Associated Sequence of Partial Sum $\{S_n\}$,
$$S_n = \sum_{k=1}^n a_k$$
 - iii. Definition of Convergence of the Infinite Series
- b. Examples of Infinite Series
 - i. Geometric Series $\sum_{n=0}^{\infty} a r^n$
 - a) Convergence for $-1 < r < 1$
 - ii. Telescoping Series
- c. General Properties
 - i. Linearity
- d. Representative Problems: 4, 7, 8, 9, 11, 12, 15, 17, 21, 23, 24, 41, 42

3. Section 8.3

- a. Divergence Test: If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ does not converge.
- b. SNNT: Series with Non-Negative Terms
 - i. Convergence Criteria: If $\sum_{n=1}^{\infty} a_n$ is SNNT and the associated sequence of partial sums is bounded above, then $\sum_{n=1}^{\infty} a_n$ converges.
- c. SNNT: Integral Test
- d. p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$: convergence for $p > 1$
- e. Representative Problems: 6, 7, 10, 12, 13, 16, 21, 24, 27, 28, 35, 37, 40, 46

4. Section 8.4

- a. SNNT: Comparison Test
 - i. Apply to Prove Convergence
 - ii. Apply to Prove Divergence
- b. SNNT: Limit Comparison Test
 - i. Apply to Prove Convergence
 - ii. Apply to Prove Divergence
- c. Representative Problems: 13, 16, 17, 20, 23, 26, 27, 31, 32, 38, 46, 48

5. Section 8.5

- a. SNNT: Ratio Test
 - i. Apply to Prove Convergence
 - ii. Apply to Prove Divergence
 - iii. Inconclusive Case
- b. SNNT: root Test
 - i. Apply to Prove Convergence
 - ii. Apply to Prove Divergence
 - iii. Inconclusive Case
- c. Representative Problems: 5, 6, 10, 14, 17, 21, 22, 29, 30, 32, 37, 40, 44

6. Section 8.6

- a. Alternating Series Test $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$
 - i. $a_n > 0$
 - ii. a_n decreasing
 - iii. $\lim_{n \rightarrow \infty} a_n = 0$
- b. Error Estimates for Approximating Alternating Series via Partial Sums
- c. Absolute Convergence
 - i. Definition
 - ii. Theorem: $\sum_{n=1}^{\infty} a_n$ converges absolutely $\Rightarrow$ $\sum_{n=1}^{\infty} a_n$ converges
- d. Generalized Ratio/Root Tests to Imply Absolute Convergence
- e. Conditional Convergence
- f. Representative Problems: 4, 7, 8, 15, 16, 20, 23, 27, 30, 32, 34, 38